

## HIRSCHENKOGEL, Austria

WMO: 113840

Lat: 47.623N

Lon: 15.833E

Elev: 4331

StdP: 12.54

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 6.6        | 10.3 | -0.9                          | 6.2 | 14.3 | 3.7 | 7.8 | 14.3 | 26.1                  | 17.8 | 23.2 | 17.3 | 7.7                   | 240  | 0.612 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 10.8                      | 73.4            | 61.5 | 70.5 | 60.6 | 67.7 | 59.2 | 63.4                | 70.5 | 61.7 | 68.5 | 60.1 | 66.4 | 4.4                     | 40   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 60.8                            | 93.5 | 67.5 | 59.0 | 87.5 | 65.7 | 57.4 | 82.6 | 63.9 | 31.2          | 70.8 | 29.8 | 68.8 | 28.6 | 66.6 | 68.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 20.7          | 16.8 | 13.6 | 4.6                        | 78.5 | 4.3                | 2.6 | 1.5                                                | 80.4 | -1.0       | 81.9 | -3.5       | 83.4 | -6.6       | 85.3 |    |    |
| (5)               |      |      | 3.9                        | 66.3 | 3.9                | 1.5 | 1.1                                                | 67.4 | -1.2       | 68.2 | -3.4       | 69.0 | -6.3       | 70.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 41.2        | 25.3  | 26.7  | 31.6  | 39.6  | 47.5  | 54.2  | 57.5  | 57.4  | 48.7  | 42.9  | 34.5  | 27.9  |  |  |
|                                                                        | DBStd       | 13.77       | 8.17  | 8.83  | 8.35  | 7.84  | 7.19  | 7.23  | 6.46  | 6.26  | 6.83  | 7.81  | 8.16  | 8.12  |  |  |
|                                                                        | HDD50       | 4001        | 767   | 651   | 571   | 320   | 135   | 39    | 12    | 10    | 104   | 241   | 467   | 684   |  |  |
|                                                                        | HDD65       | 8701        | 1232  | 1071  | 1036  | 762   | 544   | 328   | 243   | 247   | 488   | 685   | 916   | 1149  |  |  |
|                                                                        | CDD50       | 801         | 0     | 0     | 0     | 7     | 56    | 166   | 245   | 239   | 66    | 21    | 1     | 0     |  |  |
|                                                                        | CDD65       | 28          | 0     | 0     | 0     | 0     | 0     | 5     | 11    | 12    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 58          | 0     | 0     | 0     | 0     | 0     | 11    | 19    | 27    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 1           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 5.7         | 6.5   | 6.3   | 6.2   | 5.7   | 6.0   | 5.6   | 5.3   | 5.0   | 5.3   | 5.3   | 5.5   | 6.1   |  |  |
|                                                                        | PrecAvg     | 42.2        | 2.0   | 2.0   | 3.0   | 2.6   | 4.6   | 5.6   | 5.4   | 4.8   | 4.0   | 3.0   | 2.8   | 2.4   |  |  |
| (15) Precipitation                                                     | PrecMax     | 51.0        | 4.5   | 4.6   | 7.6   | 5.9   | 8.0   | 9.3   | 10.7  | 7.2   | 9.3   | 6.3   | 5.6   | 4.8   |  |  |
|                                                                        | PrecMin     | 30.4        | 0.3   | 0.3   | 1.1   | 0.2   | 1.7   | 2.4   | 1.1   | 1.4   | 1.7   | 0.5   | 0.0   | 0.4   |  |  |
|                                                                        | PrecStd     | 5.4         | 1.3   | 1.1   | 1.6   | 1.3   | 1.7   | 1.8   | 2.5   | 1.5   | 1.7   | 1.4   | 1.2   | 1.1   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 46.8  | 48.6  | 52.5  | 61.7  | 68.8  | 75.4  | 76.4  | 77.2  | 67.5  | 62.9  | 54.8  | 46.6  |  |  |
|                                                                        |             | MCWB        | 37.6  | 37.6  | 42.2  | 50.3  | 57.3  | 62.7  | 63.5  | 62.4  | 58.5  | 52.5  | 45.2  | 36.3  |  |  |
|                                                                        | 2%          | DB          | 42.7  | 44.5  | 48.5  | 56.8  | 65.2  | 71.4  | 73.0  | 73.3  | 63.5  | 58.9  | 50.7  | 43.3  |  |  |
|                                                                        |             | MCWB        | 35.4  | 35.9  | 40.5  | 47.5  | 55.3  | 60.5  | 62.1  | 61.1  | 56.4  | 51.1  | 44.3  | 36.6  |  |  |
|                                                                        | 5%          | DB          | 39.2  | 41.5  | 45.3  | 53.2  | 61.2  | 67.7  | 70.1  | 70.0  | 60.9  | 55.7  | 48.0  | 40.8  |  |  |
|                                                                        |             | MCWB        | 34.5  | 35.6  | 39.1  | 45.7  | 53.1  | 59.1  | 60.8  | 60.3  | 55.1  | 49.3  | 43.0  | 35.3  |  |  |
|                                                                        | 10%         | DB          | 36.4  | 38.2  | 42.5  | 49.9  | 58.1  | 64.7  | 67.2  | 67.0  | 58.1  | 52.9  | 45.5  | 38.5  |  |  |
|                                                                        |             | MCWB        | 32.5  | 33.6  | 37.3  | 43.6  | 51.3  | 57.5  | 59.6  | 59.2  | 53.6  | 48.2  | 41.8  | 34.1  |  |  |
|                                                                        | 0.4%        | WB          | 39.9  | 40.6  | 43.6  | 51.0  | 59.3  | 64.9  | 65.6  | 64.8  | 60.1  | 54.4  | 48.7  | 40.5  |  |  |
|                                                                        |             | MCDB        | 43.9  | 44.8  | 49.8  | 59.8  | 67.1  | 72.6  | 73.3  | 71.9  | 65.9  | 59.2  | 51.8  | 42.8  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 37.3  | 37.9  | 41.5  | 48.4  | 56.0  | 62.1  | 63.7  | 62.8  | 57.5  | 52.3  | 46.1  | 38.4  |  |  |
|                                                                        |             | MCDB        | 40.1  | 42.5  | 46.8  | 55.2  | 63.1  | 68.5  | 70.5  | 70.5  | 62.0  | 56.8  | 49.1  | 41.1  |  |  |
|                                                                        | 5%          | WB          | 35.1  | 35.9  | 39.8  | 46.3  | 53.8  | 60.0  | 61.8  | 61.3  | 55.8  | 50.4  | 43.9  | 36.6  |  |  |
|                                                                        |             | MCDB        | 38.3  | 40.0  | 44.2  | 52.0  | 60.1  | 66.5  | 68.4  | 68.3  | 59.9  | 54.0  | 46.8  | 39.6  |  |  |
|                                                                        | 10%         | WB          | 32.8  | 34.0  | 38.1  | 44.3  | 51.8  | 57.9  | 60.1  | 59.7  | 54.1  | 48.7  | 42.1  | 34.8  |  |  |
|                                                                        |             | MCDB        | 35.9  | 38.0  | 41.9  | 49.0  | 57.3  | 64.2  | 66.3  | 65.6  | 57.6  | 51.9  | 44.8  | 37.7  |  |  |
|                                                                        | 5% DB       | MDBR        | 7.1   | 8.1   | 9.0   | 10.2  | 11.1  | 11.1  | 11.4  | 10.8  | 9.2   | 8.9   | 7.2   | 7.0   |  |  |
|                                                                        |             | MCDBR       | 10.1  | 11.2  | 12.8  | 14.2  | 15.6  | 15.0  | 15.2  | 14.9  | 12.8  | 11.8  | 9.6   | 10.0  |  |  |
|                                                                        |             | MCWBR       | 6.4   | 6.8   | 7.8   | 8.1   | 8.8   | 8.5   | 7.9   | 7.8   | 8.2   | 7.1   | 6.1   | 6.3   |  |  |
|                                                                        |             | MCWBR       | 9.0   | 10.3  | 11.5  | 13.3  | 14.8  | 14.0  | 14.2  | 13.7  | 11.8  | 10.8  | 8.4   | 8.6   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.261 | 0.283 | 0.315 | 0.350 | 0.352 | 0.358 | 0.361 | 0.360 | 0.337 | 0.313 | 0.290 | 0.262 |  |  |
|                                                                        |             | taud        | 2.466 | 2.441 | 2.412 | 2.363 | 2.387 | 2.391 | 2.375 | 2.392 | 2.442 | 2.501 | 2.527 | 2.504 |  |  |
|                                                                        | Ebn at Noon |             | 267   | 283   | 288   | 286   | 289   | 287   | 284   | 280   | 277   | 267   | 253   | 254   |  |  |
|                                                                        |             | Edh at Noon | 20    | 25    | 31    | 36    | 36    | 36    | 36    | 35    | 30    | 24    | 19    | 17    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 360   | 630   | 976   | 1354  | 1582  | 1726  | 1704  | 1520  | 1095  | 725   | 396   | 291   |  |  |
|                                                                        | RadStd      |             | 38    | 80    | 113   | 140   | 173   | 130   | 150   | 164   | 135   | 71    | 49    | 39    |  |  |

## Historical Trends

|                         |  | Heating    |            | Cooling    |            |            | Degree-Days |            |            |            |            |
|-------------------------|--|------------|------------|------------|------------|------------|-------------|------------|------------|------------|------------|
|                         |  | 99% DB     | 99% DP     | 1% DB      | 1% WB      | 1% DP      | HDD50       | HDD65      | CDD50      | CDD65      |            |
| Station Only            |  | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A  | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A |
| Regional (95 neighbors) |  | +1.01      | N/A        | N/A        | +1.55      | +0.81      | +0.72       | -205       | -274       | +160       | +90        |

Nomenclature: See separate page